

PAPER 1 – END SEM / FINAL PAPER

23CSE306 – MACHINE LEARNING

Q1

Explain **Reinforcement Learning** in detail. An autonomous car uses reinforcement learning to navigate. Describe the **reward** and **environment** components in this application.

Why is proper data pre-processing, including feature transformations, essential before training a machine learning model?

Q2

OR

A large e-commerce company wants to personalize the online shopping experience but lacks explicit user feedback. Instead of data for direct recommendation, propose an **unsupervised learning** approach that could segment customers into meaningful groups based solely on their browsing history and purchase patterns (e.g., products viewed, categories clicked, purchase frequency).

Justify why unsupervised learning is appropriate and describe what insights each segment might provide.

You have 100 images, 80 of cats and 20 of dogs. If your model correctly identifies all 80 cats but misclassifies all 20 dogs as cats, calculate the **accuracy** and comment on its effectiveness.

Q3

A) Given dataset for predicting student's test score (Y) based on hours studied (X):

Hours Studied (X)	Test Score (Y)
1	50
2	55
3	65

Using the least squares method, estimate the line of best fit in the form: $Y = a + bX$

B) When would you choose a **Decision Tree** over **KNN**?

Q4

OR

Justify using **ensemble methods** instead of a single supervised model.

Analyze the use of a **Logistic Regression** model in real-life applications.

Q5

Explain the impact of **data preprocessing** on clustering.

Weights (kg) of 9 persons: **42, 45, 47, 60, 62, 65, 80, 82, 85**

Using **K = 3** and initial centroids: **C1 = 45, C2 = 60, C3 = 80**, perform one iteration of **K-Means clustering** using Euclidean distance and find:

Cluster assignments

Updated centroids

Q6

OR

Apply the **K-Means algorithm** to divide the values {2, 4, 10, 12, 3, 20, 30, 11, 25} into 2 clusters. Consider: **C1 = 4, C2 = 11**.

Design a clustering-based system for a real-world application domain.

Q7

Explain **hard voting** and **soft voting** in ensemble learning.

Analyze the improvement in accuracy through **stacked generalization** versus individual models.

Q8

OR

Evaluate the advantages and limitations of **Random Forest** compared to **Gradient Boosting Machines**.

What is **Stacked Generalization (Stacking)** in ensemble learning? Describe its components and training process.

Q9

Describe the architecture and applications of **Recurrent Neural Networks (RNNs)**. How do they handle time-dependent data?

Critically evaluate the following activation functions:

Sigmoid

Tanh

ReLU

Leaky ReLU

Compare their:

Advantages

Disadvantages

Suitability for different neural network layers

Q10

OR

Judge when to use **recurrent networks** instead of **MLP**.

Compare **Gradient Descent** and **Stochastic Gradient Descent**.